PulseCam

Contactless heart-rate measurement using only a camera | pulsecam.site

What it is

PulseCam is a web app that estimates a person's heart rate from a video of their face. There is nothing to wear, clip on or install. The user opens the page in a browser, clicks **Start camera**, sits still for about 15 seconds, and sees their pulse in beats per minute (BPM).

The problem it addresses

- **Measuring pulse needs hardware.** Pulse oximeters, chest straps and smartwatches cost money, must be owned, charged and put on, and are not always at hand.
- **Contact sensors are not always suitable.** Contact can be uncomfortable, unhygienic when shared, or impractical for infants, people with sensitive skin and people who cannot easily use their hands.
- **Heart-rate awareness is hard to teach.** Most people never see how their pulse responds to rest, breathing or exercise, and a camera-based tool makes this easy to show in a classroom or at home.

How it works

Each heartbeat pushes blood into the skin of the face, which changes its colour very slightly. The change is invisible to the eye but a camera can pick it up. This technique is called remote photoplethysmography (rPPG).

1. A face tracker (MediaPipe) finds the face in each video frame and crops it.
2. A pretrained neural network, **EfficientPhys**, turns the stream of cropped frames into a pulse signal.
3. The signal is filtered and analysed in the frequency domain to find the dominant rhythm, which is the heart rate.
4. The reading is smoothed, and is only updated when the signal quality is good enough. If the signal is weak, the number is greyed out and holds its last value.

What the app shows

Heart rate (BPM)	The main reading, with a signal-quality label and SNR.
RMSSD and SDNN	Heart-rate-variability measures, which reflect beat-to-beat changes.
Respiration	Estimated breaths per minute, when the signal allows.
Live charts	Pulse waveform (last 15 s), frequency spectrum, and heart-rate trend over the session.

Validation	The user can type in a reference reading from a pulse oximeter or chest strap, and the app shows the error and mean absolute error.

Key benefits

- **Private:** all processing happens in the user's browser. Video is never uploaded or stored.
- **Accessible:** works on a laptop or phone with an ordinary webcam. No app, account or sensor.
- **Free and instant:** a web link or a QR code is all that is needed.
- **Transparent:** the built-in validation panel lets anyone check the accuracy against a real device.

Tips for a good reading

- Sit about 50 cm from the camera, with the device propped up rather than held.
- Face a bright, steady, even light source.
- Stay still and keep still for about 15 seconds while the reading settles.

Limitations

PulseCam is a school project and **not a medical device**. It must not be used to diagnose or monitor any medical condition. Accuracy drops with movement, poor or flickering light, and heavy facial occlusion. Skin tone, camera quality and compression can also affect results.

Technology and credits

EfficientPhys model and weights from the rPPG-Toolbox (Liu et al., WACV 2023 and NeurIPS 2023), trained on the PURE dataset; MediaPipe Face Landmarker for face tracking; ONNX Runtime Web for in-browser inference. Hosted on GitHub Pages.